Umurinzi Cyber

Explainable Machine Learning Intrusion Detection for Rwandan institutions.

The Problem

Cybersecurity Problem

- Cyber attacks increasing in Rwanda as the digital economy grows.
- African SMEs cannot afford enterprise IDS solutions in developed countries where even 72% have no IT Security staff.
- Traditional IDS detect only known attacks and with no reason specified that analyst can act on.

Effects

- Financial losses
- Data breaches
- Service disruption
- Reduced trust

Goal of Project

Main Goal

Develop an Explainable AI-based IDS that:

- Detects cyber attacks
- Explains predictions
- Supports non-expert analysts

Attack Classes

- Normal
- DoS
- Probe
- Exploitation
- Backdoor

Why This Solution

Why machine Learning?

- Learns attack patterns
- Detects unseen behavior
- Better accuracy

Why Explainable AI?

- Analysts need trust
- Understand why alerts occur
- Better decisions

Why Explainable AI?

- No hardware dependency
- More Accessible
- Less Costs

Data

Each row = one network conversation (flow) summarised as statistics: how many packets were sent, how fast, how large, what TCP flags. It's basically the measurable traces every attack leaves behind.

CICIDS2017

Created by Simulated Institutional network with attacks including DDoS, DoS, Port Scanning, Brute Force, Web Attacks, Botnet activity, and Infiltration.

UNSW-NB15

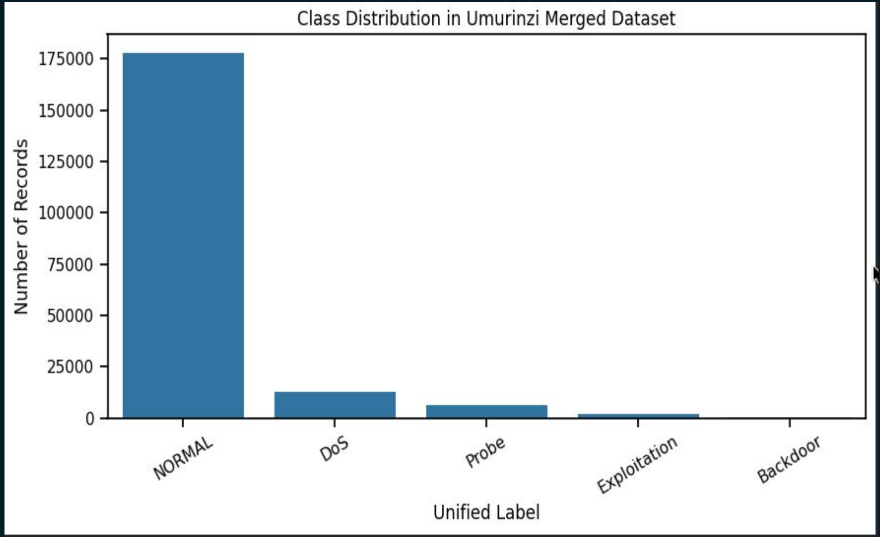
Generated using IXIA traffic generator in a real network testbed.

The data was merged by first removing duplicate records, missing values, and infinite values to improve data quality. The attack labels were then mapped into five categories: **NORMAL**, **DoS**, **Probe**, **Exploitation**, and **Backdoor**. Finally, feature alignment was performed to create a unified schema between the **CICIDS2017** and **UNSW-NB15** datasets.

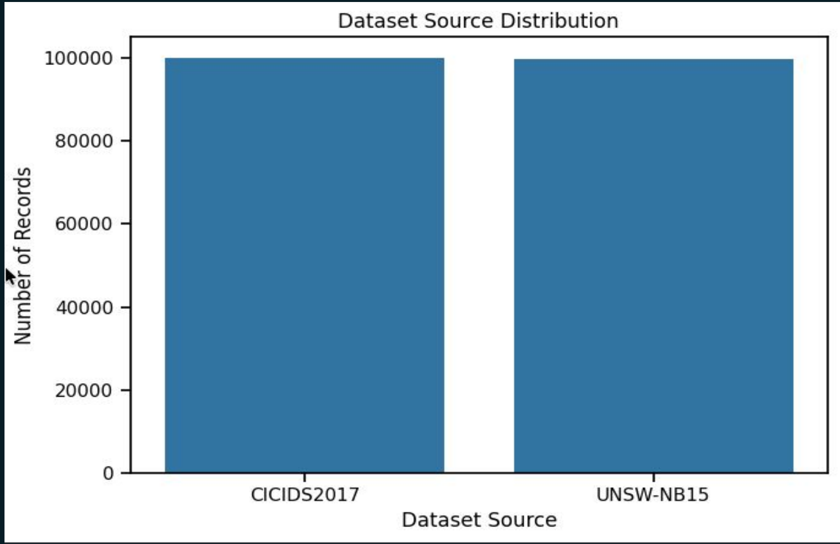
80% data: Training
20% data: Test

Data Exploration: Class & Source Distribution

Dataset is heavily imbalance with NORMAL class dominating (88%) requires oversampling (SMOTE).

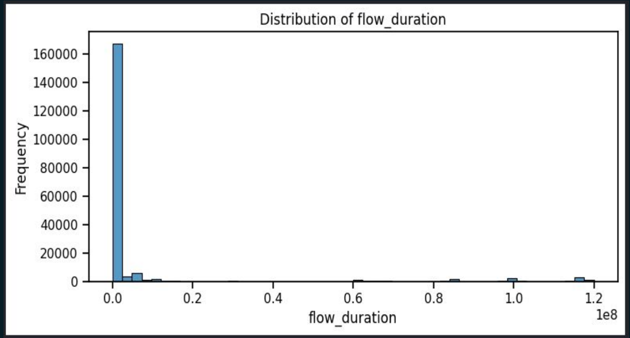


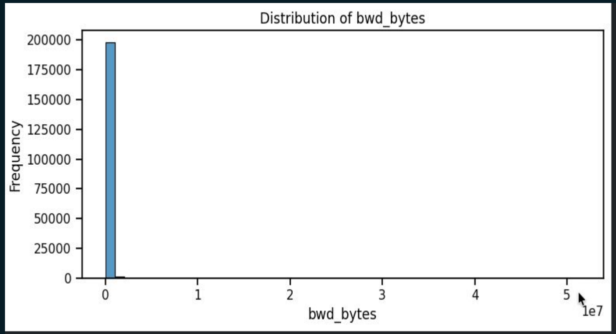
Source balance — equal split: 100k records each from CICIDS2017 and UNSW-NB15

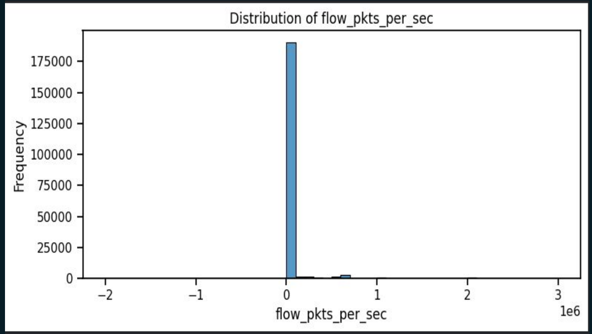


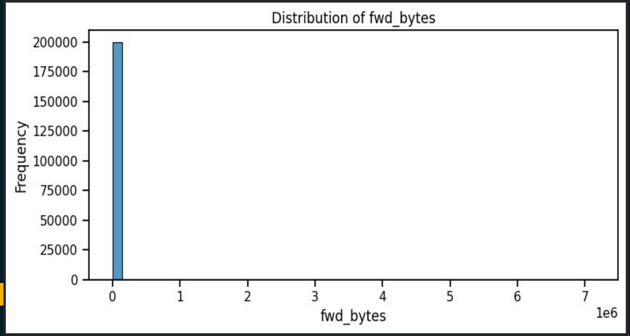
Feature Distributions

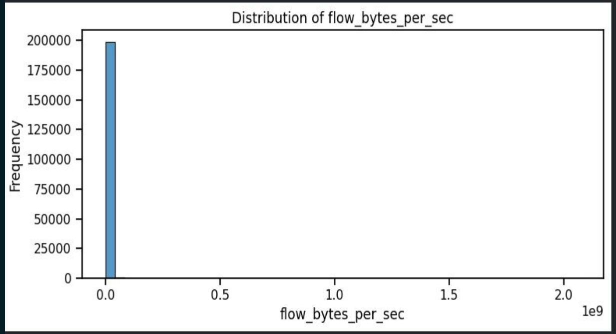
- Most network connections have small values (short duration, few packets, little data transfer), while only a small number of connections have very large values.
- Most network traffic is directed to a small range of common ports, such as those used by web servers, email services, and other standard applications that are already frequently used port numbers.

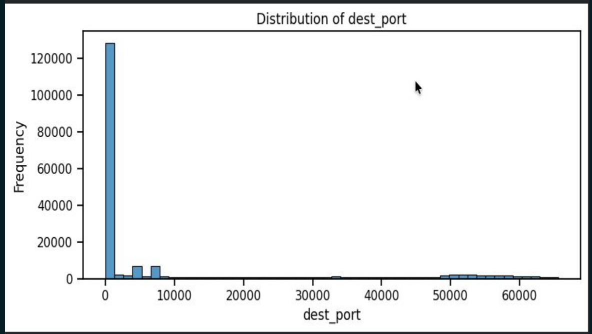






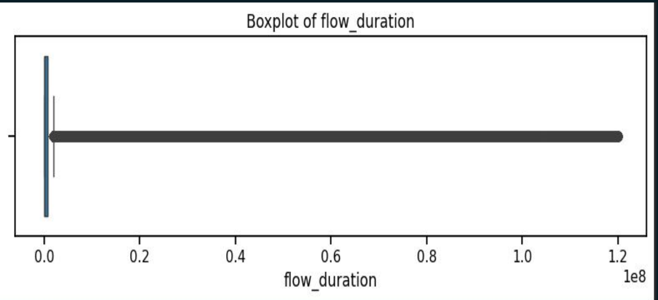


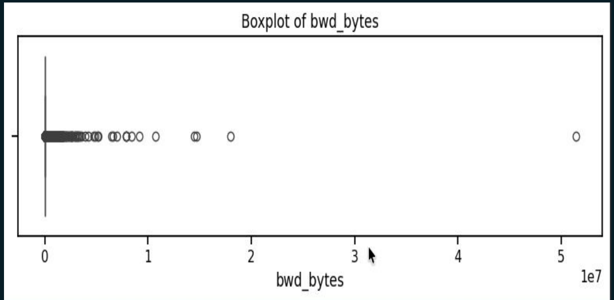


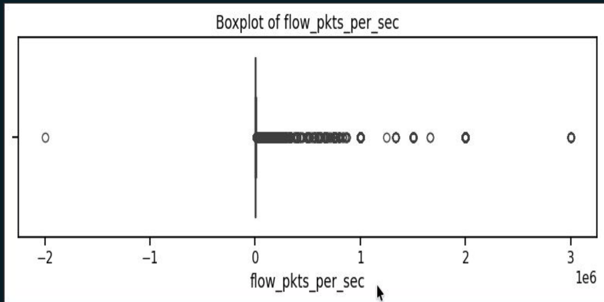


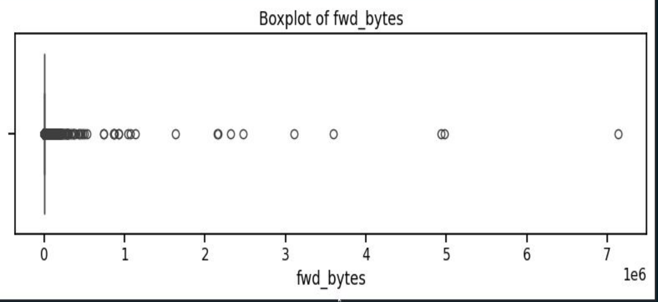
Outlier Analysis

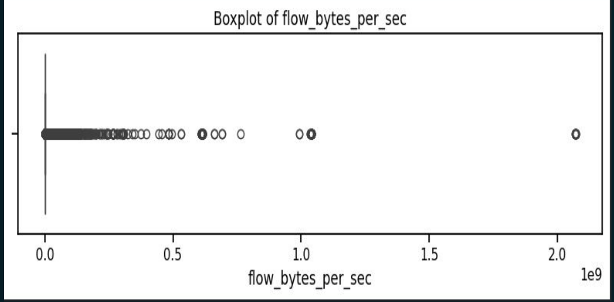
- Some network flows have **much larger byte counts and traffic rates** than the majority of the data.
- The *flow_duration* feature varies the most, with some connections lasting **much longer than others**, which may be associated with backdoor activity.
- The outliers were **not removed** because they could contain important information about actual cyberattacks and abnormal network behavior.

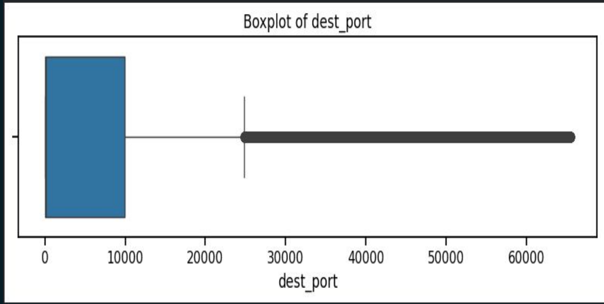






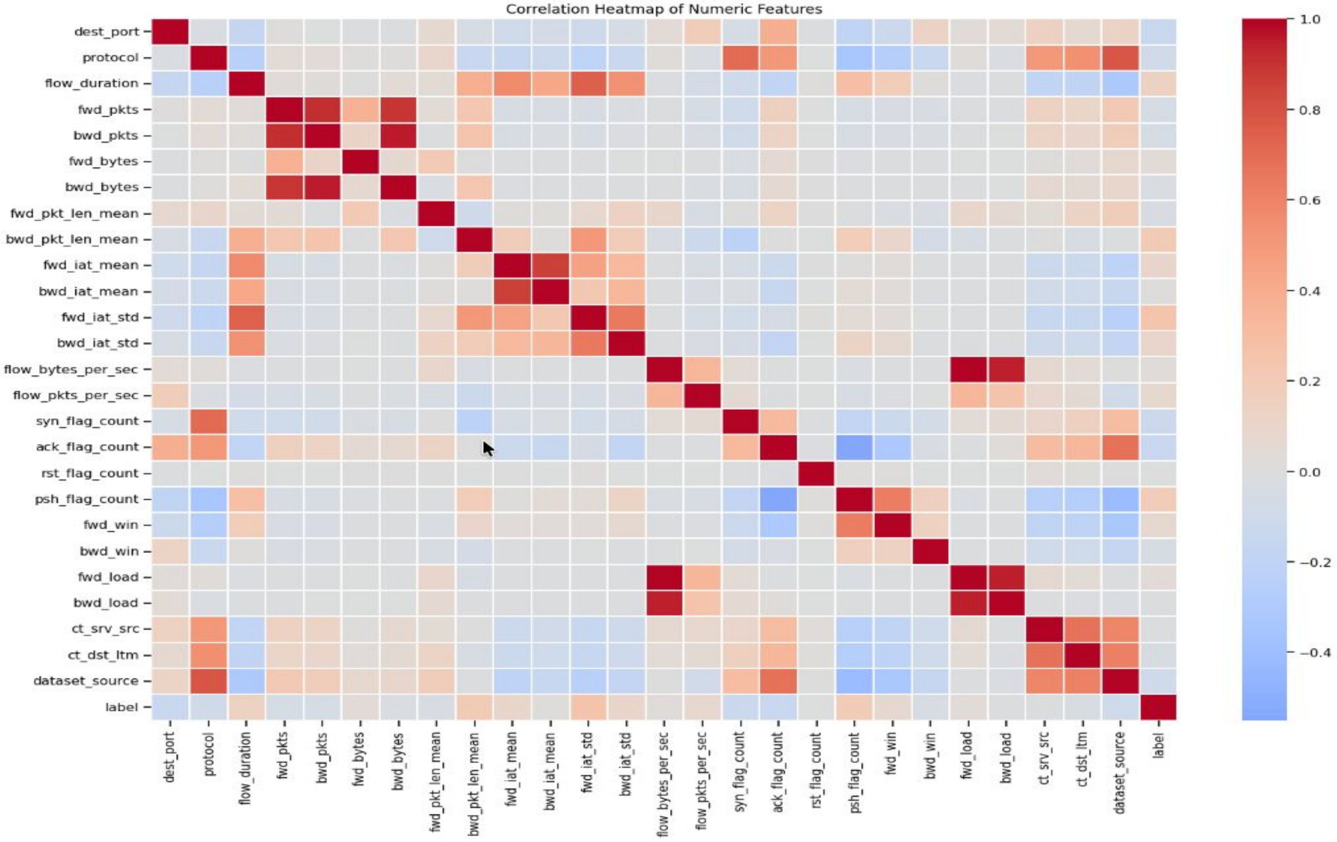






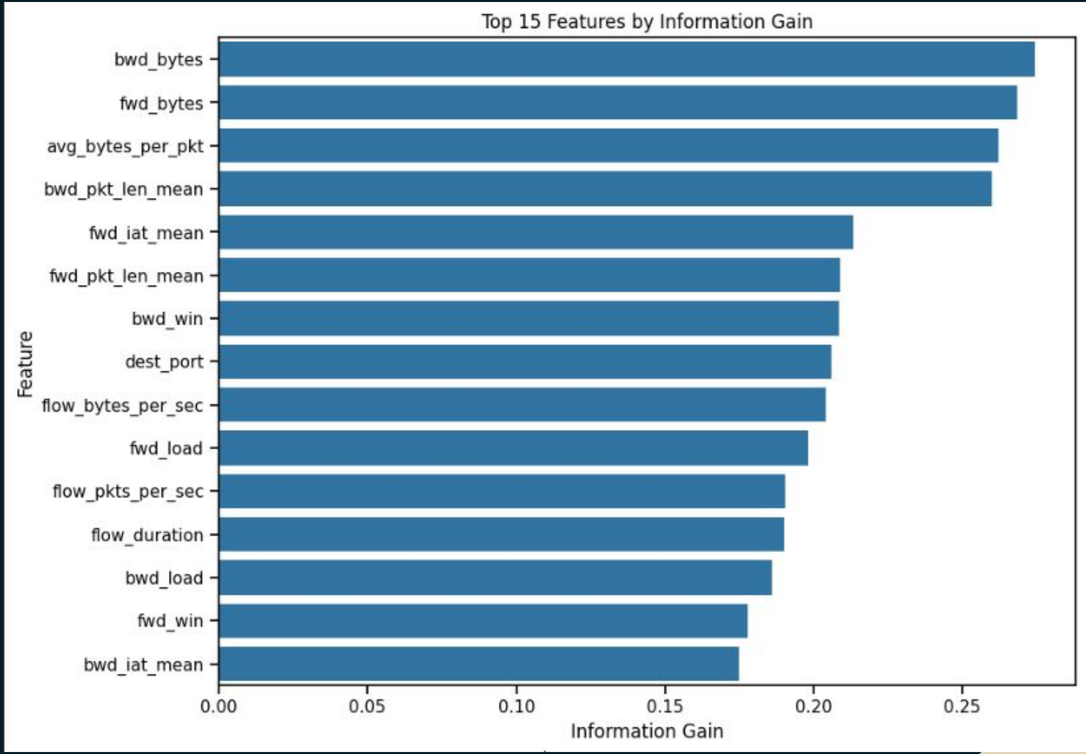
Correlation

The number of bytes sent is closely related to the number of packets sent. Meaning when a network flow contains more packets, it usually also contains **more data (bytes)**



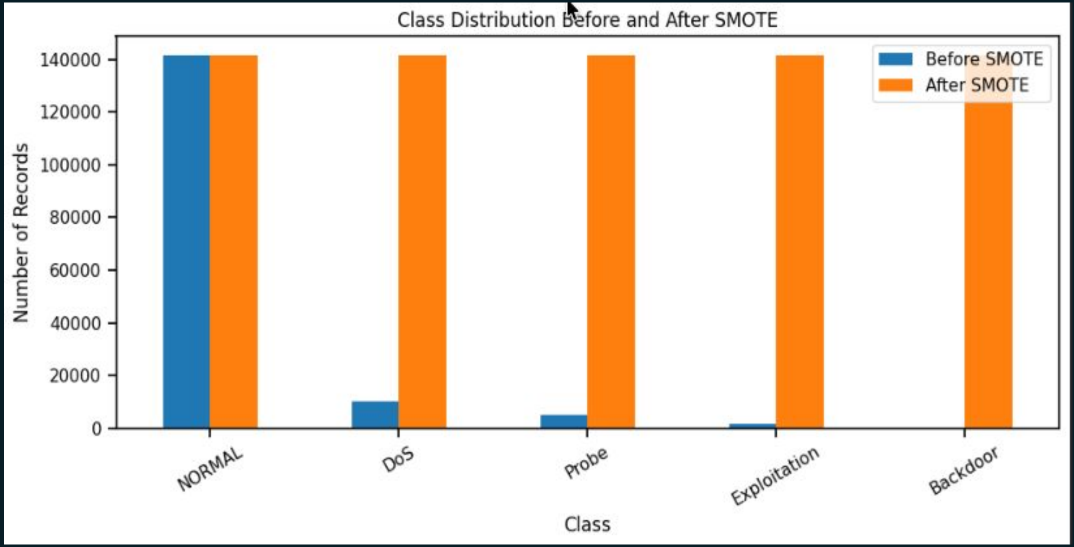
Feature Selection

- Mutual information was used to measure how useful each feature is for predicting the target class (normal traffic or attack type).
- Top features: bwd_bytes, fwd_bytes, avg_bytes_per_pkt, bwd_pkt_len_mean
- Features that contributed little useful information were removed.



Handling Class Imbalance — SMOTE

- SMOTE was used to generate new synthetic examples for classes that had fewer samples in the training data.
- Each minority class was increased until it contained approximately 140,000 samples, making the class distribution more balanced.
- SMOTE was applied only to the training set to help the model learn from all classes equally.
- The test set was left unchanged so that model performance could be evaluated on data that reflects the real-world class distribution.



Model Building

1
80/20 Split

2
3 Models

3
Tune

5-fold cross-validation + grid search for hyperparameters

4
Evaluate

Accuracy, F1, Precision, Recall, False Positive Rate

5
Best Model

Highest F1-score selected for deployment

The Models

Logistic Regression

Simple Linear Classifier. Expected to be weak on complex, imbalance traffic data.

Random Forest

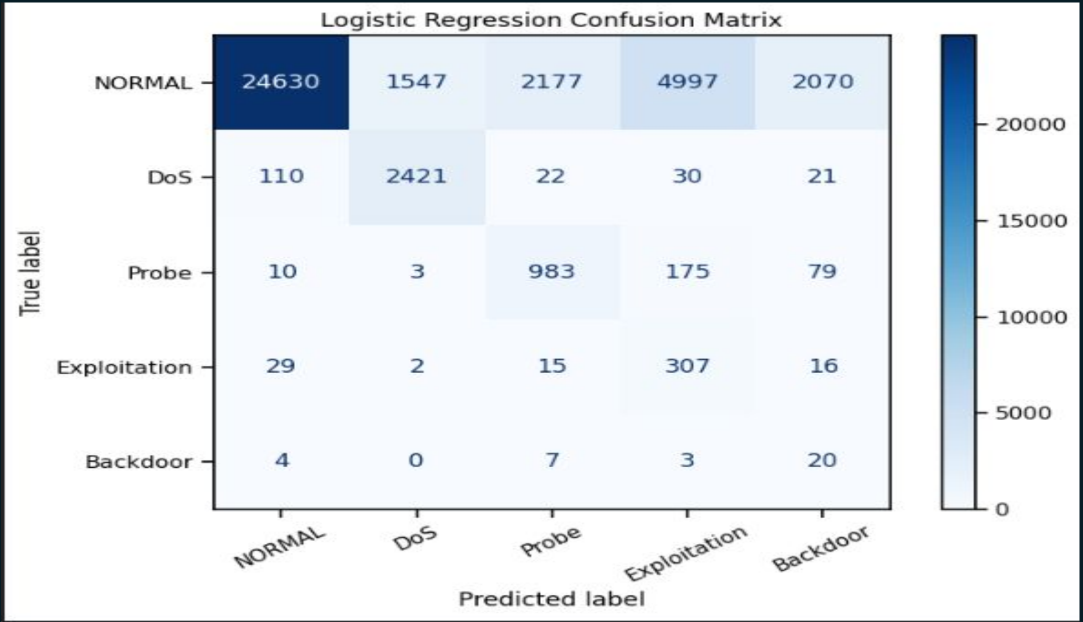
Less affected by noisy, irrelevant, or unusual data and naturally compatible with SHAP.

XGBoost

XGBoost achieved the highest performance among the tested models, making it the strongest candidate for the final intrusion detection system.

Model 1: Logistic Regression (Baseline Results)

Metric	Value
Accuracy	71.48%
Precision	39.50%
Recall	76.63%
F1-Score	42.36%
False Positive Rate	6.46%



- The model correctly identified a large proportion of **Normal** and **DoS** traffic
- Performance dropped significantly for **Exploitation** and **Backdoor** attacks because these classes contain very few examples and their traffic patterns overlap with other attack categories.
- The confusion matrix shows that many Normal records were incorrectly classified as attacks, which reduced overall precision.

Discussion (LR Results)

The Logistic Regression model achieved 71.48% accuracy and 76.63% recall. This means the model was able to detect most attack instances, which is important in cybersecurity because missing an attack can be costly.

However, the precision was relatively low at 39.5%, indicating that many benign records were incorrectly flagged as attacks.

The confusion matrix shows that the model performs well for Normal and DoS traffic but struggles with minority classes such as Backdoor and Exploitation. This behavior is expected because Logistic Regression assumes linear relationships, whereas network intrusion patterns are often highly non-linear.

Therefore, Logistic Regression serves as our baseline model, and the next stage will evaluate Random Forest and XGBoost to determine whether more sophisticated models can improve performance.

Conclusion & Future Work

Conclusion

- ❑ Merged CICIDS2017 + UNSW-NB15
- ❑ Created unified dataset
- ❑ Completed EDA
- ❑ Trained Logistic Regression baseline

Remaining Work

Remaining Work

- ❑ Random Forest
- ❑ XGBoost
- ❑ SHAP Explainability
- ❑ Isolation Forest for unknown attack detection
- ❑ FastAPI backend
- ❑ React dashboard
- ❑ User study